

Complete-Propagator Composite-Gate Hierarchies from Identical Phase-Shifted Pulses

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Fully compensating and phase-preserving composite rotations are well established for pulse-area, detuning, and combined systematic errors. Here we address a more specific problem: order-by-order cancellation of the *complete* propagator when every constituent pulse is represented by the same, otherwise unknown, $SU(2)$ matrix and only its programmed phase is changed. For nominal π constituents, parity separates diagonal and off-diagonal corrections and organizes analytic five- and nine-pulse branches together with higher-order numerical $X = R_x(\pi)$ gates. For nominal $\pi/2$ constituents, a bivariate coefficient recursion generates fixed $R_x(\pi/2)$ gates: nine pulses cancel all first-order terms, eighteen unrestricted or nineteen palindromic pulses cancel every second-order term, and a palindromic 31-pulse sequence cancels all terms through total third order. We additionally construct selective higher-order families in which prescribed mixed cubic terms are removed. Thus the main contribution is not the first realization of a robust constant rotation, but a unified identical-constituent Cayley–Klein hierarchy for complete-propagator cancellation in two abstract error coordinates. Complete numerical phase lists are reported in the appendices, and phase-sensitive validation protocols are proposed for experimental tests.

I. INTRODUCTION

Composite pulses replace one imperfect rotation by a programmed sequence whose systematic errors interfere destructively. The subject originated in nuclear magnetic resonance with broadband inversion and self-compensating excitation sequences [1–4]. Importantly, the literature has long distinguished population transfer from a fully compensating, or class-A, rotation that acts correctly on an arbitrary input state. Phase-preserving composite π and $\pi/2$ pulses were demonstrated already in the early work on self-compensating and phase-distortionless rotations [2, 5].

Constant or fully compensating rotations have subsequently been developed for many target angles and error models. The Wimperis broadband, narrowband, and passband families include arbitrary-angle rotations robust to pulse-strength errors [6]; BB1, SCROFULOUS, CORPSE, and related sequences were formulated explicitly as complete single-qubit gates [7, 8]. High-order π and numerical $\pi/2$ families were studied in Ref. [9], while arbitrary-order and equiangular response-polynomial methods provide systematic constructions for complete rotations [10–12]. More recent work has constructed ultrahigh-fidelity X , Hadamard, arbitrary-angle, narrowband, and passband rotational gates [13, 14]. Detuning errors can be treated by CORPSE-type and more general symmetric sequences [7, 15], whereas concatenated constructions can compensate pulse-length and off-resonance errors simultaneously [16]. Therefore, the existence of robust constant rotations, including π and $\pi/2$ gates, is not itself new.

A complementary notion of universality removes the need to specify the microscopic error Hamiltonian. Each constituent is represented by the same unknown $SU(2)$ Cayley–Klein propagator, while the experimentally programmed phases are the design variables. This approach

suppresses reproducible errors without assigning them separately to pulse-area offsets, detuning, pulse-shape distortions, chirp, or other physical mechanisms. It led to universal composite pulses for population inversion [17, 18] and to broadband, narrowband, passband, and variable-rotation protocols [19–22]. These constructions are experimentally attractive because every pulse envelope may remain identical and only the phase reference changes.

For a quantum gate, however, robust population transfer alone is insufficient. A population-inversion sequence constrains the magnitude of a transition amplitude but may leave its phase, and hence the rotation axis, dependent on the unknown constituent propagator. Here we use *constant rotation* in the same full-unitary sense as fully compensating or phase-distortionless control: the complete composite propagator must approach a prescribed unitary matrix, up to global phase. The distinction must be tested using phase-sensitive observables, process tomography, or gate benchmarking rather than population measurements alone [23–27].

A recent preliminary study by two of the present authors constructed full-unitary X and half-rotation gates under explicit amplitude, detuning, and duration-error models [28]. In particular, it identified the two five-pulse X branches as phase-shifted forms of the universal-inversion sequences $U5a$ and $U5b$. The present work does not claim those low-order constant rotations as new. Instead, it reformulates the problem at the level of an otherwise unknown identical constituent propagator and develops a complete two-coordinate cancellation hierarchy. This exposes the parity structure of nominal π constituents, provides a common full-propagator classification of analytic and numerical X sequences, and, most substantially, produces unrestricted and palindromic $R_x(\pi/2)$ families with complete second- and third-order cancella-

tion, together with selective cancellation of prescribed cubic monomials.

The paper is organized as follows. We first define the fixed-axis Cayley–Klein model and distinguish complete-propagator cancellation from a population profile. We then present the analytic five- and nine-pulse X branches as the low-order members of the parity hierarchy and collect all longer numerical X solutions in Appendix B. The main $R_x(\pi/2)$ section presents the multivariate coefficient recursion and resource counting, while the numerical phase vectors and verification details are collected in Appendices D and E.

II. UNIVERSAL FIXED-AXIS MODEL

For either nominal rotation, a programmed phase ϕ changes only the azimuthal phase of the off-diagonal Cayley–Klein element. A common additional phase β obeys

$$U_\phi(\beta) = Z(\beta)U_\phi(0)Z^\dagger(\beta), \quad (1a)$$

$$Z(\beta) = e^{i\beta\sigma_z/2}. \quad (1b)$$

Since the same β occurs in every constituent,

$$U_{\text{CP}}(\beta) = Z(\beta)U_{\text{CP}}(0)Z^\dagger(\beta). \quad (2)$$

Thus phase-only concatenation cannot produce a laboratory-fixed equatorial axis that is independent of an arbitrary unknown β . We therefore set $\beta = 0$ as the controlled drive-phase reference and require universality with respect to the remaining common coordinates. Throughout, we use $R_x(\theta) = \exp(-i\theta\sigma_x/2)$.

Writing

$$U_{\text{CP}} = \begin{pmatrix} A & B \\ -B^* & A^* \end{pmatrix}, \quad (3)$$

a variable rotation constrains only a probability such as $|B|^2$, whereas a constant rotation requires

$$U_{\text{CP}} = U_t + \text{higher-order terms}, \quad (4)$$

with $U_t = R_x(\pi)$ or $R_x(\pi/2)$. Population measurements alone therefore cannot establish constant-rotation behavior.

Throughout this work, fidelity denotes the average gate fidelity with respect to the target unitary. For a qubit channel $\mathcal{E}$ it is defined by

$$F_{\text{avg}}(\mathcal{E}, U_t) = \int d\psi \langle \psi | U_t^\dagger \mathcal{E}(|\psi\rangle\langle\psi|) U_t | \psi \rangle, \quad (5)$$

where the integral is taken over the uniform Haar measure on pure qubit states. For the unitary composite propagators considered in the theoretical analysis, Eq. (5) reduces to

$$F_{\text{avg}}(U_{\text{CP}}, U_t) = \frac{|\text{Tr}(U_t^\dagger U_{\text{CP}})|^2 + 2}{6}. \quad (6)$$

The quantity plotted below is the average gate infidelity $1 - F_{\text{avg}}$. This definition is insensitive to a global phase and equals unity when the implemented and target gates agree up to such a phase.

III. CONSTANT UNIVERSAL X GATES

For an identical nominal π pulse with programmed phase ϕ , we use

$$U_\phi^{(\pi)}(q, \alpha) = \begin{pmatrix} qe^{i\alpha} & -i\sqrt{1-q^2}e^{i\phi} \\ -i\sqrt{1-q^2}e^{-i\phi} & qe^{-i\alpha} \end{pmatrix}, \quad (7)$$

where q is the single-pulse failure amplitude and α is an arbitrary common dynamical phase. For every odd-length sequence, the Cayley–Klein elements have opposite parity,

$$A(q, \alpha) = qA_1 + q^3A_3 + q^5A_5 + \dots, \quad (8a)$$

$$B(q, \alpha) = B_0 + q^2B_2 + q^4B_4 + q^6B_6 + \dots. \quad (8b)$$

A proof of this parity structure is given in Appendix F. Consequently, successive orders alternate between diagonal and off-diagonal cancellation conditions.

A. Analytic five-pulse gates

For the palindromic ansatz $\Phi = (\phi_1, \phi_2, \phi_3, \phi_2, \phi_1)$, the ideal-target condition and $A_1(\alpha) = 0$ for every α produce two inequivalent branches. Choosing all phases modulo 2π , they may be written as

$$\Phi_{X5a} = \left(\frac{2\pi}{3}, \frac{11\pi}{6}, \frac{\pi}{3}, \frac{11\pi}{6}, \frac{2\pi}{3} \right) \quad (9a)$$

$$\Phi_{X5b} = \left(\frac{2\pi}{3}, \frac{5\pi}{6}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{2\pi}{3} \right). \quad (9b)$$

Both branches satisfy

$$U_{\text{CP}}^{(X5a)} = U_{\text{CP}}^{(X5b)} = R_x(\pi) + \mathcal{O}(q^2), \quad 1 - F_{\text{avg}} = \mathcal{O}(q^4). \quad (10)$$

They obey the same cancellation conditions but have different leading second-order off-diagonal coefficients and therefore distribute the finite- q error differently over α . After a common phase shift, these two branches coincide with the universal-inversion sequences $U5a$ and $U5b$ of Ref. [17]; their full-unitary interpretation under explicit amplitude, detuning, and duration errors was given in Ref. [28]. We retain them here as the analytic base of the unknown-constituent parity hierarchy. Phase conjugation gives a symmetry-equivalent partner for each branch; the complete classification is summarized in Appendix B.

B. Analytic nine-pulse gates

For the palindromic phase string

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4, \phi_5, \phi_4, \phi_3, \phi_2, \phi_1), \quad \phi_1 = 0, \quad (11)$$

the universal conditions $A_1 = B_2 = 0$ reduce to the relations

$$\phi_4 = \phi_2 + \pi, \quad (12a)$$

$$\phi_3 = 2\phi_2 - \phi_5 + \pi, \quad (12b)$$

$$e^{2i\phi_5} - e^{i\phi_5} + 1 = 0, \quad (12c)$$

$$2e^{4i\phi_2} + (1 + e^{i\phi_5})e^{2i\phi_2} + 2e^{i\phi_5} = 0 \quad (12d)$$

The central phase follows directly from Eq. (12c). With $z_5 = e^{i\phi_5}$, one obtains

$$z_5^2 - z_5 + 1 = 0, \quad (13a)$$

$$z_5 = e^{\pm i\pi/3}, \quad \phi_5 = \pm \frac{\pi}{3} \pmod{2\pi}. \quad (13b)$$

The two signs are related by phase conjugation of the complete sequence, so we take $\phi_5 = \pi/3$ without loss of an inequivalent branch. Defining

$$\vartheta = \arccos\left(-\frac{\sqrt{3}}{4}\right), \quad (14)$$

the two roots of the quadratic phase equation produce two inequivalent constant- X branches. Applying the common phase shift $\chi = \pi/3$ fixes the target axis to x and gives

$$\Phi_{X9a} = \left(\frac{\pi}{3}, \frac{5\pi}{12} - \frac{\vartheta}{2}, \frac{7\pi}{6} - \vartheta, \frac{17\pi}{12} - \frac{\vartheta}{2}, \frac{2\pi}{3}, \right. \\ \left. \frac{17\pi}{12} - \frac{\vartheta}{2}, \frac{7\pi}{6} - \vartheta, \frac{5\pi}{12} - \frac{\vartheta}{2}, \frac{\pi}{3} \right) \quad (15a)$$

$$\Phi_{X9b} = \left(\frac{\pi}{3}, \frac{5\pi}{12} + \frac{\vartheta}{2}, \frac{7\pi}{6} + \vartheta, \frac{17\pi}{12} + \frac{\vartheta}{2}, \frac{2\pi}{3}, \right. \\ \left. \frac{17\pi}{12} + \frac{\vartheta}{2}, \frac{7\pi}{6} + \vartheta, \frac{5\pi}{12} + \frac{\vartheta}{2}, \frac{\pi}{3} \right). \quad (15b)$$

Both sequences obey

$$U_{\text{CP}}^{(X9a)} = U_{\text{CP}}^{(X9b)} = R_x(\pi) + \mathcal{O}(q^3), \quad 1 - F_{\text{avg}} = \mathcal{O}(q^6). \quad (16)$$

As for $X5$, the cancellation order is identical while the higher-order coefficients, and hence the finite-error fidelity landscapes, distinguish the two branches. Related longer full-unitary X sequences have been optimized in specific amplitude-detuning models [28]; the role of the present analytic $X9$ solutions is to exhibit the next parity step directly in the abstract (q, α) constituent coordinates. Their phase-conjugate partners are symmetry-equivalent.

Higher-order constant-rotation solutions at $N = 17, 23, 25, 35$, and 41 are numerical. Their complete phase vectors and cancellation conditions are listed in Table I of Appendix B. The Genov $U13$ sequences instead optimize universal population inversion: they cancel A_1 and A_3 but do not generally cancel the gate-phase correction B_2 , and therefore are not members of the constant-rotation hierarchy. Rectangular-pulse amplitude-detuning landscapes for the complete Table I catalog are shown in Fig. 1.

IV. CONSTANT UNIVERSAL H GATES

We use H as shorthand for the fixed half rotation $R_x(\pi/2)$. One nominal constituent is parameterized as

$$U_{\phi}^{(\pi/2)}(\epsilon, \alpha) = \begin{pmatrix} \sqrt{\frac{1}{2} - \epsilon} e^{i\alpha} & -i\sqrt{\frac{1}{2} + \epsilon} e^{i\phi} \\ -i\sqrt{\frac{1}{2} + \epsilon} e^{-i\phi} & \sqrt{\frac{1}{2} - \epsilon} e^{-i\alpha} \end{pmatrix}. \quad (17)$$

The complete propagator is expanded in the two universal coordinates,

$$U_{\text{CP}} = \sum_{m,n \geq 0} \epsilon^m \alpha^n U_{mn}, \quad U_{00} = R_x(\pi/2). \quad (18)$$

For each pulse, write similarly $U_{\phi_k}^{(\pi/2)} = \sum_{m,n} \epsilon^m \alpha^n V_{mn}(\phi_k)$. If $C_{mn}^{(k)}$ denotes the coefficient after the first k pulses, direct multiplication gives the convolution recursion

$$C_{mn}^{(k)} = \sum_{r=0}^m \sum_{s=0}^n V_{rs}(\phi_k) C_{m-r, n-s}^{(k-1)}, \quad (19)$$

with $C_{00}^{(0)} = I$ and all other initial coefficients zero. This is the construction principle: choose the phases so that $C_{00}^{(N)} = R_x(\pi/2)$ and set the desired collection of higher coefficients to zero.

The successive complete-cancellation conditions are

$$U_{10} = U_{01} = 0, \quad (20a)$$

$$U_{20} = U_{11} = U_{02} = 0, \quad (20b)$$

$$U_{30} = U_{21} = U_{12} = U_{03} = 0. \quad (20c)$$

Equation (20a) gives complete first-order compensation; Eq. (20b) is additionally imposed for complete second-order compensation, and Eq. (20c) completes the conditions through total third order. An unrestricted fixed $SU(2)$ target and five vanishing first- and second-order coefficients generically provide $3+5 \times 3 = 18$ real constraints, explaining the observed unrestricted threshold $N = 18$. Palindromic symmetry changes the independent-condition count and produces a complete second-order solution at $N = 19$. Selective cubic cancellation yields palindromic families at $N = 21$ – 29 , while the $H31a$ sequence cancels every term through total third order,

$$U_{\text{CP}}^{(H31a)} = R_x(\pi/2) + \mathcal{O}(\rho^4), \quad 1 - F_{\text{avg}} = \mathcal{O}(\rho^8), \quad (21)$$

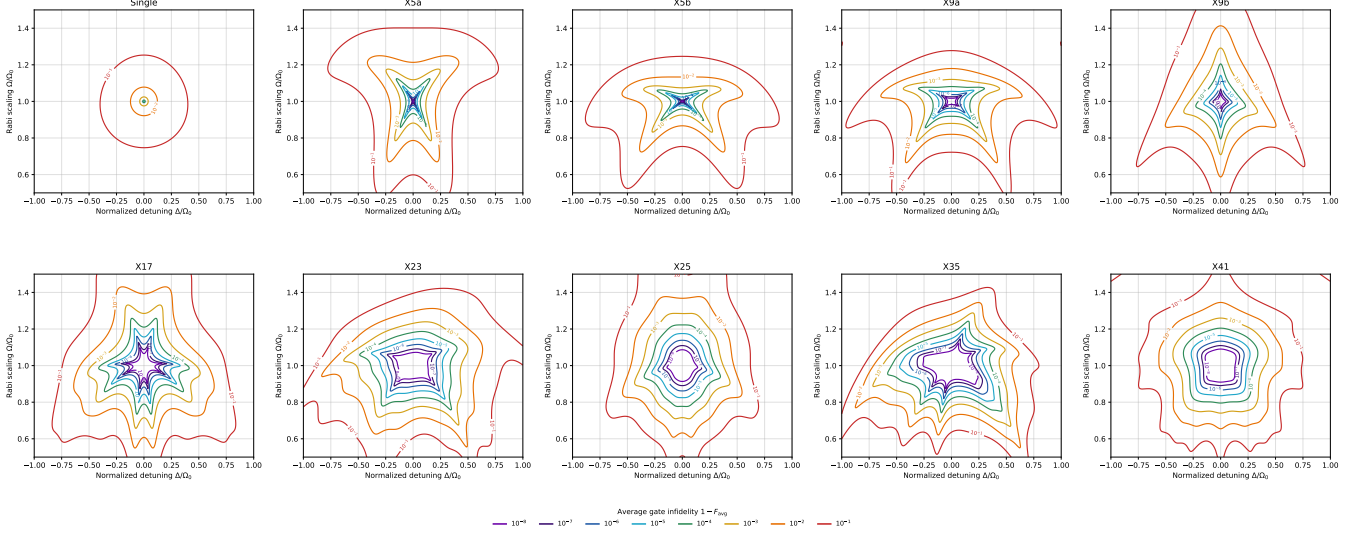


FIG. 1. Average gate infidelity landscapes for the constant $X = R_x(\pi)$ gates in Table I, evaluated for identical rectangular constituent pulses. The top row shows the single pulse, $X5a$, $X5b$, $X9a$, and $X9b$; the bottom row shows $X17$, $X23$, $X25$, $X35$, and $X41$. In every panel the horizontal axis is the normalized detuning Δ/Ω_0 , the vertical axis is the Rabi scaling Ω/Ω_0 , and the nominal pulse satisfies $\Omega_0 T = \pi$. Contours mark $1 - F_{\text{avg}} = 10^{-8}, 10^{-7}, \dots, 10^{-1}$, with the same color assigned to each level in all panels.

with ρ a generic small displacement in the (ϵ, α) plane. All numerical phase vectors, including unrestricted and palindromic branches, are collected in Table II of Appendix E. A comparison of selected rectangular-pulse H -gate landscapes is given in Fig. 2.

V. EXPERIMENTAL IMPLEMENTATION AND VALIDATION PROTOCOLS

The constituent-pulse assumption is directly implementable whenever a platform supports phase-programmable rotations with otherwise identical envelopes. On a superconducting processor one may use $\text{PRX}(\theta, \phi_k)$ pulses with a common amplitude scaling and a common drive-frequency offset applied to every segment. The same protocol applies to optical or microwave ion-qubit rotations by changing only the carrier phase.

A convincing demonstration must be phase sensitive. We propose four linked measurements. First, reconstruct the complete process matrix under a two-dimensional scan of amplitude and detuning errors and report $1 - F_{\text{avg}}$, not only the excited-state probability. Second, extract the effective rotation angle and Bloch-sphere axis from unitary or process tomography; the azimuthal axis should remain fixed for a constant sequence but drift for a variable-rotation control. Third, use a Ramsey interferometer in which the composite gate is placed between phase-referenced analysis pulses; this directly resolves the transition phase. Fourth, perform interleaved randomized benchmarking or cycle benchmarking under controlled coherent offsets to separate the predicted robustness from

SPAM errors.

For a compact first experimental paper, the most informative gate set is

$$X : \{1, 5, 9, 17\} \text{ pulses,} \quad (22a)$$

$$H : \{1, 9, 18\} \text{ pulses.} \quad (22b)$$

with $N = 23$ and 35 added only if coherence permits. The experiment should also show the duration-error tradeoff: at very small coherent error, decoherence and phase noise can make a shorter sequence outperform a formally higher-order one.

A future benchmarking figure should compare the perturbative slopes near the calibrated point with the predicted orders and display the crossover at which decoherence makes the longer sequences disadvantageous.

VI. CONCLUSION

Fully compensating composite rotations have a substantial history, ranging from phase-distortionless π and $\pi/2$ pulses to arbitrary-angle gates robust to amplitude, detuning, and combined errors. The contribution of the present work is therefore not the introduction of constant rotations themselves. Rather, we formulate a common hierarchy for phase-only control when every constituent is the same otherwise unknown $SU(2)$ propagator and robustness is imposed on the complete unitary in two abstract Cayley–Klein coordinates.

Within this formulation, the X -gate problem has a parity-separated expansion. The analytic $N = 5$ and

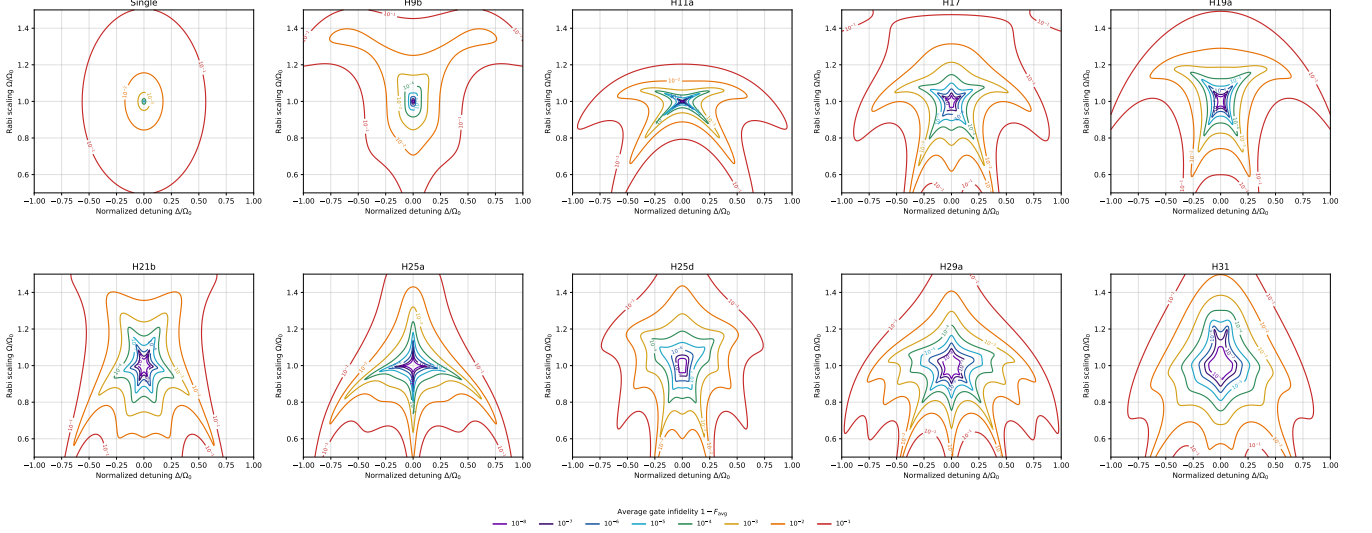


FIG. 2. Average gate infidelity landscapes for representative constant $H = R_x(\pi/2)$ gates, evaluated for identical rectangular constituent pulses. The top row shows the single pulse, $H9b$, $H11a$, the selective palindromic $H17$ sequence, and $H19a$; the bottom row shows $H21b$, $H25a$, $H25d$, $H29a$, and $H31$. In every panel the horizontal axis is the normalized detuning Δ/Ω_0 , the vertical axis is the Rabi scaling Ω/Ω_0 , and the nominal pulse satisfies $\Omega_0 T = \pi/2$. Contours mark $1 - F_{\text{avg}} = 10^{-8}, 10^{-7}, \dots, 10^{-1}$, with a common level-to-color mapping across all panels.

$N = 9$ branches serve as the first members of a hierarchy whose longer solutions cancel alternating diagonal and off-diagonal propagator coefficients. The half-rotation problem is genuinely multivariate and is generated systematically by the coefficient recursion (19). Its principal new output is the catalog of unrestricted and palindromic identical- $\pi/2$ sequences with complete second- and third-order cancellation, including families that selectively remove specified mixed cubic terms. The remaining open tasks are exhaustive minimality searches, algebraic reconstruction of selected numerical roots, and phase-sensitive hardware validation of the full propagator.

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Appendix A: Common phase covariance and target convention

Restoring the common off-diagonal phase, the primitive and composite propagators obey

$$U_\phi(\beta) = Z(\beta)U_\phi(0)Z^\dagger(\beta), \quad (\text{A1a})$$

$$U_{\text{CP}}(\beta) = U_{\phi_N}(\beta) \cdots U_{\phi_1}(\beta) = Z(\beta)U_{\text{CP}}(0)Z^\dagger(\beta). \quad (\text{A1b})$$

Here the phases $\phi_1, \dots, \phi_N$ are ordered chronologically. Therefore no choice of the programmed phases can eliminate an arbitrary common β . A common programmed shift $\phi_k \mapsto \phi_k + \chi$ preserves the cancellation order but conjugates the target by a known Z rotation, thereby changing the prescribed equatorial axis.

Appendix B: X-gate derivation and complete phase catalog

For odd N , every path contributing to the diagonal Cayley–Klein element contains an odd number of diagonal primitive factors, while every path contributing to the off-diagonal element contains an even number. Since the diagonal primitive factor is proportional to q and the off-diagonal magnitude $\sqrt{1 - q^2}$ is even in q , Eq. (8) follows. Every coefficient $A_{2r+1}(\alpha)$ or $B_{2r}(\alpha)$ is a finite Laurent polynomial in $z = e^{i\alpha}$; universality requires every Laurent coefficient to vanish separately.

1. Five- and nine-pulse analytic constructions

For the five-pulse palindrome, introduce $t_k = e^{i\phi_k}$. The target condition and the independent Laurent coefficients

of $A_1(\alpha)$ reduce to

$$\frac{t_1^2 t_3}{t_2^2} = 1, \quad (\text{B1a})$$

$$\frac{t_2}{t_3} + \frac{t_3}{t_2} = 0, \quad (\text{B1b})$$

$$\frac{t_1 t_3}{t_2^2} + 1 + \frac{t_2^2}{t_1 t_3} = 0. \quad (\text{B1c})$$

The roots separate into two inequivalent branches, represented by Eqs. (9a) and (9b). Their phase-conjugate partners are

$$-\Phi_{X5a} \pmod{2\pi} = \left(\frac{4\pi}{3}, \frac{\pi}{6}, \frac{5\pi}{3}, \frac{\pi}{6}, \frac{4\pi}{3} \right), \quad (\text{B2a})$$

$$-\Phi_{X5b} \pmod{2\pi} = \left(\frac{4\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3} \right). \quad (\text{B2b})$$

Thus the four algebraic solutions form two inequivalent conjugate pairs.

For the palindromic nine-pulse phase string

$$\Phi = (\phi_1, \phi_2, \phi_3, \phi_4, \phi_5, \phi_4, \phi_3, \phi_2, \phi_1), \quad \phi_1 = 0, \quad (\text{B3})$$

the conditions $A_1 = B_2 = 0$ reduce to

$$\phi_4 = \phi_2 + \pi, \quad (\text{B4a})$$

$$\phi_3 = 2\phi_2 - \phi_5 + \pi, \quad (\text{B4b})$$

$$e^{2i\phi_5} - e^{i\phi_5} + 1 = 0, \quad (\text{B4c})$$

$$2e^{4i\phi_2} + (1 + e^{i\phi_5}) e^{2i\phi_2} + 2e^{i\phi_5} = 0. \quad (\text{B4d})$$

whose roots are

$$z_5^2 - z_5 + 1 = 0, \quad (\text{B5a})$$

$$z_5 = e^{\pm i\pi/3}, \quad \phi_5 = \pm \frac{\pi}{3} \pmod{2\pi}. \quad (\text{B5b})$$

The two signs generate phase-conjugate sequences and do not define additional inequivalent branches. We therefore choose $\phi_5 = \pi/3$. Let $\xi = e^{2i\phi_2}$ and $\omega = e^{i\pi/3}$. The remaining equation and its roots can be written as

$$2\xi^2 + (1 + \omega)\xi + 2\omega = 0, \quad (\text{B6a})$$

$$\xi = e^{i(\pi/6 \pm \vartheta)}, \quad (\text{B6b})$$

$$\vartheta = \arccos\left(-\frac{\sqrt{3}}{4}\right). \quad (\text{B6c})$$

After the common phase shift $\chi = \pi/3$ fixes the ideal axis to x , these roots give the two inequivalent sequences in Eqs. (15a) and (15b). Both obey $U_{CP} = R_x(\pi) + \mathcal{O}(q^3)$ for every α . Their phase conjugates give two additional algebraic representatives, so the nine-pulse solutions likewise form two inequivalent conjugate pairs.

The $N = 17$ list is palindromic and cancels A_1, B_2, A_3 . The unrestricted $N = 23$ list additionally cancels B_4 , while $N = 25$ is a palindromic representative at the same order. The unrestricted $N = 35$ sequence cancels A_1, B_2, A_3, B_4, A_5 , and $N = 41$ is the corresponding palindromic representative. Laurent-polynomial extraction from the full-precision $N = 35$ phases, before rounding for presentation, gives maximum residuals 7.5×10^{-16} , 2.6×10^{-14} , 3.1×10^{-14} , 9.4×10^{-14} , 1.9×10^{-13} , and 4.5×10^{-13} for $B_0 + i$, A_1, B_2, A_3, B_4 , and A_5 , respectively.

2. Complete constant-rotation X -gate phase vectors

All phases in Table I are written in chronological order

Writing $z_5 = e^{i\phi_5}$, Eq. (B4c) becomes a quadratic equation and are divided by π .
 TABLE I: Constant-rotation $R_x(\pi)$ solutions and their phase vectors. All entries in the final column are divided by π , with $\vartheta = \arccos(-\sqrt{3}/4)$.

X gate	Cancelled terms	Φ/π
Single	none	(0)
$X5a$	A_1	$\left(\frac{2}{3}, \frac{11}{6}, \frac{1}{3}, \frac{11}{6}, \frac{2}{3}\right)$
$X5b$	A_1	$\left(\frac{2}{3}, \frac{5}{6}, \frac{1}{3}, \frac{5}{6}, \frac{2}{3}\right)$
$X9a$	A_1, B_2	$\left(\frac{1}{3}, \frac{5}{12} - \frac{\vartheta}{2\pi}, \frac{7}{6} - \frac{\vartheta}{\pi}, \frac{17}{12} - \frac{\vartheta}{2\pi}, \frac{2}{3}, \frac{17}{12} - \frac{\vartheta}{2\pi}, \frac{7}{6} - \frac{\vartheta}{\pi}, \frac{5}{12} - \frac{\vartheta}{2\pi}, \frac{1}{3}\right)$
$X9b$	A_1, B_2	$\left(\frac{1}{3}, \frac{5}{12} + \frac{\vartheta}{2\pi}, \frac{7}{6} + \frac{\vartheta}{\pi}, \frac{17}{12} + \frac{\vartheta}{2\pi}, \frac{2}{3}, \frac{17}{12} + \frac{\vartheta}{2\pi}, \frac{7}{6} + \frac{\vartheta}{\pi}, \frac{5}{12} + \frac{\vartheta}{2\pi}, \frac{1}{3}\right)$
$X17$	A_1, B_2, A_3	(1.325072, 0.439152, 0.410726, 1.438278, 1.753746, 1.040188, 0.097953, 0.105859, 0.871961, 0.105859, 0.097953, 1.040188, 1.753746, 1.438278, 0.410726, 0.439152, 1.325072)
$X23$	A_1, B_2, A_3, B_4	(0.312203, 0.323342, 0.657051, 0.564744, 1.522338, 0.979449, 1.569721, 1.603010, 1.206396, 0.392728, 1.320471, 1.847060, 1.679493, 1.061946, 1.558854, 1.150341, 1.942482, 1.776578, 0.911067, 0.295065, 0.090991, 0.394749, 0.617944)
$X25$	A_1, B_2, A_3, B_4	(0.644304, 1.714057, 1.516593, 0.887803, 0.819587, 1.771161, 1.519103, 0.031881, 1.254160, 1.258790, 0.261978, 1.903485, 1.102906, 1.903485, 0.261978, 1.258790, 1.254160, 0.031881, 1.519103, 1.771161, 0.819587, 0.887803, 1.516593, 1.714057, 0.644304)
$X35$	A_1, B_2, A_3, B_4, A_5	(0.169018, 0.403633, 0.742774, 0.529058, 1.442517, 1.727390, 1.521617, 0.718591, 1.066878, 0.872412, 0.410098, 1.379533, 1.705300, 1.500478, 0.462491, 0.620874, 0.367432, 0.971952, 0.788349, 1.645179, 0.770531, 0.909429, 1.785802, 1.600373, 0.028027, 1.727407, 0.108171, 0.148086, 1.894671, 1.179833, 1.989955, 0.322637, 0.036950, 1.120357, 1.086641)

Continued on next page

TABLE I (continued)

X gate	Cancelled terms	Φ/π
X41	A_1, B_2, A_3, B_4, A_5	(0.224567, 1.125864, 1.927495, 1.426323, 0.041898, 1.934501, 1.206162, 1.471487, 0.636987, 1.386301, 1.076870, 1.748411, 1.031420, 1.396435, 0.534009, 0.272628, 0.808716, 0.006958, 1.791659, 1.261540, 1.501327, 1.261540, 1.791659, 0.006958, 0.808716, 0.272628, 0.534009, 1.396435, 1.031420, 1.748411, 1.076870, 1.386301, 0.636987, 1.471487, 1.206162, 1.934501, 0.041898, 1.426323, 1.927495, 1.125864, 0.224567)

3. Relation to the Genov $U13$ inversion sequences

The palindromic universal-inversion sequences of Ref. [17] are

$$\frac{\Phi_{U13a}}{\pi} = \frac{1}{24}(0, 9, 42, 11, 8, 37, 2, 37, 8, 11, 42, 9, 0), \quad (\text{B7a})$$

$$\frac{\Phi_{U13b}}{\pi} = \frac{1}{24}(0, 33, 42, 35, 8, 13, 2, 13, 8, 35, 42, 33, 0). \quad (\text{B7b})$$

They satisfy $A_1 = A_3 = 0$, hence $1 - P = \mathcal{O}(q^{10})$, but generally retain $B_2 \neq 0$. After a common target-axis shift, their full propagator error is therefore only $\mathcal{O}(q^2)$. This contrasts with the five-pulse members $U5a$ and $U5b$, whose phase-shifted forms do satisfy the complete first-order gate conditions [28]. The $U13$ sequences are included here for comparison but not in Table I, which contains constant-rotation solutions only.

Appendix C: Numerical verification

Two independent checks were used. First, direct products were expanded in the parity-compatible form

$$A = qA_1 + q^3A_3 + q^5A_5 + \dots, \quad (\text{C1a})$$

$$B = B_0 + q^2B_2 + q^4B_4 + q^6B_6 + \dots. \quad (\text{C1b})$$

Second, the quantity

$$E_N(q, \alpha) = \|U_{\text{CP}}(q, \alpha) - U_{\text{CP}}(0, \alpha)\|_{\text{F}} \quad (\text{C2})$$

was evaluated on logarithmic q grids. The fitted slopes approach 2, 3, 4, 5, and 6 for $N = 5, 9, 17, 23$, and 35, respectively.

Appendix D: H -gate coefficient construction

Let $s = 1/\sqrt{2}$. Through total degree two, Eq. (17) is

$$U_{\phi}^{(\pi/2)} = V_0 + \epsilon V_{\epsilon} + \alpha V_{\alpha} + \epsilon^2 V_{\epsilon^2} + \epsilon \alpha V_{\epsilon\alpha} + \alpha^2 V_{\alpha^2} + \mathcal{O}_3, \quad (\text{D1})$$

where $\mathcal{O}_3$ collects all monomials of total degree three and higher, and

$$V_0(\phi) = s \begin{pmatrix} 1 & -ie^{i\phi} \\ -ie^{-i\phi} & 1 \end{pmatrix}, \quad (\text{D2a})$$

$$V_{\epsilon}(\phi) = s \begin{pmatrix} -1 & -ie^{i\phi} \\ -ie^{-i\phi} & -1 \end{pmatrix}, \quad (\text{D2b})$$

$$V_{\alpha}(\phi) = is \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (\text{D2c})$$

$$V_{\epsilon^2}(\phi) = \frac{s}{2} \begin{pmatrix} -1 & ie^{i\phi} \\ ie^{-i\phi} & -1 \end{pmatrix}, \quad (\text{D2d})$$

$$V_{\epsilon\alpha}(\phi) = -is \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (\text{D2e})$$

$$V_{\alpha^2}(\phi) = -\frac{s}{2}I. \quad (\text{D2f})$$

Define the partial product

$$Q^{(k)} = C_0^{(k)} + \epsilon C_{\epsilon}^{(k)} + \alpha C_{\alpha}^{(k)} + \epsilon^2 C_{\epsilon^2}^{(k)} + \epsilon \alpha C_{\epsilon\alpha}^{(k)} + \alpha^2 C_{\alpha^2}^{(k)} + \mathcal{O}_3. \quad (\text{D3})$$

Here $C_0^{(0)} = I$ and all other initial coefficients zero. Multiplication by pulse k gives

$$C_0^{(k)} = V_0(\phi_k)C_0^{(k-1)}, \quad (\text{D4a})$$

$$C_{\epsilon}^{(k)} = V_0C_{\epsilon}^{(k-1)} + V_{\epsilon}C_0^{(k-1)}, \quad (\text{D4b})$$

$$C_{\alpha}^{(k)} = V_0C_{\alpha}^{(k-1)} + V_{\alpha}C_0^{(k-1)}, \quad (\text{D4c})$$

$$C_{\epsilon^2}^{(k)} = V_0C_{\epsilon^2}^{(k-1)} + V_{\epsilon}C_{\epsilon}^{(k-1)} + V_{\epsilon^2}C_0^{(k-1)}, \quad (\text{D4d})$$

$$C_{\epsilon\alpha}^{(k)} = V_0C_{\epsilon\alpha}^{(k-1)} + V_{\epsilon}C_{\alpha}^{(k-1)} + V_{\alpha}C_{\epsilon}^{(k-1)} + V_{\epsilon\alpha}C_0^{(k-1)}, \quad (\text{D4e})$$

$$C_{\alpha^2}^{(k)} = V_0C_{\alpha^2}^{(k-1)} + V_{\alpha}C_{\alpha}^{(k-1)} + V_{\alpha^2}C_0^{(k-1)}. \quad (\text{D4f})$$

In Eqs. (D4), each V is evaluated at ϕ_k . After the final pulse, the coefficient conventions in the appendix and main text are related by

$$U_{00} = C_0^{(N)}, \quad (\text{D5a})$$

$$U_{10} = C_{\epsilon}^{(N)}, \quad U_{01} = C_{\alpha}^{(N)}, \quad (\text{D5b})$$

$$U_{20} = C_{\epsilon^2}^{(N)}, \quad U_{11} = C_{\epsilon\alpha}^{(N)}, \quad U_{02} = C_{\alpha^2}^{(N)}. \quad (\text{D5c})$$

The fixed-target conditions stated in the main text then follow directly.

A fixed $SU(2)$ target imposes three real constraints. Once the ideal target is fixed, each additional vanishing tangent-space coefficient contributes three more

real conditions. For cancellation of the five coefficients $U_{10}, U_{01}, U_{20}, U_{11}, U_{02}$, the natural unrestricted threshold is therefore $3 + 5 \times 3 = 18$ phases. This count supports, but does not prove, the observed numerical threshold.

The three $N = 18$ solutions cancel every coefficient through total second order. For the full-precision numerical roots, the residual norms lie between approximately 10^{-12} and 5×10^{-11} . The phase vectors tabulated below are rounded to six digits after the decimal point for presentation.

Appendix E: Complete numerical H -gate phase catalog

The following table contains every numerical $R_x(\pi/2)$ phase vector used in this work. Phases are written in chronological order and divided by π . The second column lists the cancelled Taylor terms. Letters distinguish inequivalent sequences having the same pulse number.

TABLE II: Complete numerical phase catalog for constant-rotation $R_x(\pi/2)$ solutions. The second column lists the cancelled Taylor monomials, with different perturbative orders placed on separate lines. Letters distinguish inequivalent sequences at fixed pulse number. All phases in the final column are divided by π .

H gate	Cancelled terms	Φ/π
Single	none	(0)
$H9a$	ϵ, α	(0.424089, 0.646413, 0.431718, 0.801954, 1.684846, 1.389272, 1.586807, 1.395014, 0.080868)
$H9b$	ϵ, α	(1.139702, 1.456655, 0.327072, 0.106809, 1.740400, 0.106809, 0.327072, 1.456655, 1.139702)
$H11a$	ϵ, α	(0.777941, 0.518699, 0.660394, 0.877660, 0.525243, 0.273513, 0.525243, 0.877660, 0.660394,
	$\epsilon\alpha$	0.518699, 0.777941)
$H11b$	ϵ, α	(0.911978, 0.518529, 0.748442, 0.880732, 0.598166, 0.266825, 0.598166, 0.880732, 0.748442,
	$\epsilon\alpha$	0.518529, 0.911978)
$H12a$	ϵ, α	(0.953958, 1.377589, 1.782202, 0.232340, 1.801920, 1.645993, 0.160805, 1.936757, 1.281550,
	α^2	0.930038, 1.100638, 1.708464)
$H12b$	ϵ, α	(0.826013, 1.640413, 1.328575, 0.346280, 0.181015, 0.279296, 0.571950, 1.669375, 1.412830,
	ϵ^2	1.811087, 0.091342, 0.873349)
$H18a$	ϵ, α	(1.410931, 1.686714, 1.287738, 0.853009, 1.325273, 0.679895, 1.870309, 0.250063, 0.021448,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	0.659642, 0.595158, 1.995421, 0.149592, 0.279189, 1.315249, 1.040226, 0.179509, 1.468625)
$H18b$	ϵ, α	(1.255940, 0.471744, 0.507937, 1.077188, 0.138672, 0.094174, 1.721287, 1.722604, 0.352383,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	0.522631, 1.822920, 1.879625, 0.137586, 0.527210, 1.298196, 1.043948, 0.699340, 0.525190)
$H18c$	ϵ, α	(0.475949, 0.211408, 0.944623, 0.831259, 0.695360, 1.628574, 1.901908, 1.890262, 1.255372,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	1.551332, 0.213612, 1.553317, 1.705824, 1.719493, 0.737064, 0.452711, 0.977968, 0.180519)
$H19a$	ϵ, α	(0.362127, 1.116678, 1.129203, 0.792324, 0.285443, 0.625713, 1.890168, 1.868427, 0.525435,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	0.449107, 0.525435, 1.868427, 1.890168, 0.625713, 0.285443, 0.792324, 1.129203, 1.116678,
		0.362127)
$H19b$	ϵ, α	(1.576524, 0.932968, 0.652583, 1.459557, 1.244279, 1.907413, 1.884880, 0.406962, 1.726487,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	1.492946, 1.726487, 0.406962, 1.884880, 1.907413, 1.244279, 1.459557, 0.652583, 0.932968,
		1.576524)
$H19c$	ϵ, α	(0.269330, 0.229157, 0.851760, 0.760520, 1.878742, 1.301280, 0.226938, 1.800282, 1.321724,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	1.703518, 1.321724, 1.800282, 0.226938, 1.301280, 1.878742, 0.760520, 0.851760, 0.229157,
		0.269330)
$H21a$	ϵ, α	(1.205662, 1.434294, 0.207236, 1.734269, 1.488467, 1.192926, 0.559625, 1.723416, 0.099241,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	0.519532, 0.169647, 0.519532, 0.099241, 1.723416, 0.559625, 1.192926, 1.488467, 1.734269,
	α^3	0.207236, 1.434294, 1.205662)
$H21b$	ϵ, α	(0.564630, 0.413663, 1.204587, 1.176848, 1.189079, 0.223200, 1.911951, 0.520870, 1.881477,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	1.531393, 1.850297, 1.531393, 1.881477, 0.520870, 1.911951, 0.223200, 1.189079, 1.176848,
	ϵ^2, α^2	1.204587, 0.413663, 0.564630)
$H23a$	ϵ, α	(0.296593, 0.684571, 1.285404, 0.624706, 0.049357, 1.509842, 0.704515, 1.454589, 0.093044,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	0.269654, 1.536008, 1.646602, 1.536008, 0.269654, 0.093044, 1.454589, 0.704515, 1.509842,
	ϵ^3	0.049357, 0.624706, 1.285404, 0.684571, 0.296593)
$H23b$	ϵ, α	(1.797652, 1.630663, 0.918300, 1.173336, 1.737274, 1.107215, 0.307500, 0.563901, 1.828269,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	0.234515, 0.811839, 0.383782, 0.811839, 0.234515, 1.828269, 0.563901, 0.307500, 1.107215,
	ϵ^2, α^2	1.737274, 1.173336, 0.918300, 1.630663, 1.797652)
$H23c$	ϵ, α	(0.614118, 1.430566, 1.246946, 0.777867, 0.589005, 0.308063, 1.303548, 0.156781, 1.442834,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	0.092258, 0.432699, 0.343736, 0.432699, 0.092258, 1.442834, 0.156781, 1.303548, 0.308063,
	$\epsilon^3, \epsilon^2, \alpha^2$	0.589005, 0.777867, 1.246946, 1.430566, 0.614118)
$H25a$	ϵ, α	(1.163781, 1.744204, 1.163465, 0.265945, 0.435447, 1.746295, 1.540338, 1.816104, 1.550579,
	$\epsilon^2, \alpha^2, \epsilon\alpha$	0.579137, 0.634461, 0.626526, 0.073539, 0.626526, 0.634461, 0.579137, 1.550579, 1.816104,
	ϵ^3, α^3	1.540338, 1.746295, 0.435447, 0.265945, 1.163465, 1.744204, 1.163781)

Continued on next page

TABLE II (continued)

H gate	Cancelled terms	Φ/π
$H25b$	$\epsilon_2 \alpha$ $\epsilon_3^2 \alpha^2, \epsilon \alpha$ $\alpha^3, \epsilon \alpha^2$	(0.815774, 1.816485, 1.487536, 0.803205, 0.772029, 1.100639, 0.297076, 1.481224, 0.219188, 1.803689, 1.857041, 0.563897, 0.142116, 0.563897, 1.857041, 1.803689, 0.219188, 1.481224, 0.297076, 1.100639, 0.772029, 0.803205, 1.487536, 1.816485, 0.815774)
$H25c$	$\epsilon_2 \alpha$ $\epsilon_3^2 \alpha^2, \epsilon \alpha$ $\epsilon, \epsilon^2 \alpha$	(1.408785, 0.046692, 1.708497, 0.110998, 1.970295, 1.777949, 0.880992, 0.196387, 1.103470, 0.954166, 1.230015, 0.482369, 0.655905, 0.482369, 1.230015, 0.954166, 1.103470, 0.196387, 0.880992, 1.777949, 1.970295, 0.110998, 1.708497, 0.046692, 1.408785)
$H25d$	$\epsilon_2 \alpha$ $\epsilon_3^2 \alpha^2, \epsilon \alpha$ $\epsilon^2 \alpha, \epsilon \alpha^2$	(1.198768, 0.131154, 1.553381, 0.904513, 0.847540, 1.250045, 0.432300, 1.647570, 0.341503, 1.835302, 1.954239, 0.661291, 0.175761, 0.661291, 1.954239, 1.835302, 0.341503, 1.647570, 0.432300, 1.250045, 0.847540, 0.904513, 1.553381, 0.131154, 1.198768)
$H27a$	$\epsilon_2 \alpha$ $\epsilon_3^2 \alpha^2, \epsilon \alpha$ $\epsilon, \epsilon \alpha^2$	(0.726908, 1.685503, 0.364393, 0.851603, 1.234571, 0.445661, 1.987847, 1.591379, 0.765472, 1.394976, 0.067724, 0.434151, 1.742585, 1.698913, 1.742585, 0.434151, 0.067724, 1.394976, 0.765472, 1.591379, 1.987847, 0.445661, 1.234571, 0.851603, 0.364393, 1.685503, 0.726908)
$H27b$	$\epsilon_2 \alpha$ $\epsilon_3^2 \alpha^2, \epsilon \alpha$ $\alpha^3, \epsilon^2 \alpha, \epsilon \alpha^2$	(0.835132, 0.025311, 0.898471, 1.541023, 1.224426, 0.741481, 0.611543, 0.492564, 1.552070, 0.187262, 1.415607, 0.124818, 0.429507, 0.424657, 0.429507, 0.124818, 1.415607, 0.187262, 1.552070, 0.492564, 0.611543, 0.741481, 1.224426, 1.541023, 0.898471, 0.025311, 0.835132)
$H27c$	$\epsilon_2 \alpha$ $\epsilon_3^2 \alpha^2, \epsilon \alpha$ $\epsilon, \alpha^3, \epsilon^2 \alpha$	(1.502803, 0.659205, 0.696133, 0.641221, 0.996956, 1.842886, 0.020492, 1.455687, 0.789033, 1.776628, 1.332689, 0.009673, 0.247762, 1.742378, 0.247762, 0.009673, 1.332689, 1.776628, 0.789033, 1.455687, 0.020492, 1.842886, 0.996956, 0.641221, 0.696133, 0.659205, 1.502803)
$H29a$	$\epsilon_2 \alpha$ $\epsilon_3^2 \alpha^2, \epsilon \alpha$ $\epsilon, \epsilon^2 \alpha, \epsilon \alpha^2$	(1.622560, 1.588167, 0.295670, 1.108906, 0.438853, 0.223129, 0.069164, 0.188553, 0.403101, 0.927562, 1.974731, 1.444300, 1.280259, 1.184073, 1.220492, 1.184073, 1.280259, 1.444300, 1.974731, 0.927562, 0.403101, 0.188553, 0.069164, 0.223129, 0.438853, 1.108906, 0.295670, 1.588167, 1.622560)
$H29b$	$\epsilon_2 \alpha$ $\epsilon_3^2 \alpha^2, \epsilon \alpha$ $\epsilon, \alpha^3, \epsilon \alpha^2$	(1.328488, 1.479406, 0.093160, 1.158666, 0.592630, 1.852256, 0.640681, 1.989136, 1.649702, 1.856274, 1.613740, 0.629669, 0.692759, 0.756344, 0.109543, 0.756344, 0.692759, 0.629669, 1.613740, 1.856274, 1.649702, 1.989136, 0.640681, 1.852256, 0.592630, 1.158666, 0.093160, 1.479406, 1.328488)
$H31a$	$\epsilon_2 \alpha$ $\epsilon_3^2 \alpha^2, \epsilon \alpha$ $\epsilon, \alpha^3, \epsilon^2 \alpha, \epsilon \alpha^2$	(0.687984, 1.619887, 1.076903, 1.336783, 0.707793, 0.602841, 0.703561, 1.007987, 0.057400, 1.727043, 1.867103, 1.633585, 0.377056, 0.391767, 0.375257, 1.789730, 0.375257, 0.391767, 0.377056, 1.633585, 1.867103, 1.727043, 0.057400, 1.007987, 0.703561, 0.602841, 0.707793, 1.336783, 1.076903, 1.619887, 0.687984)

Appendix F: Parity structure of odd-length X -gate sequences

In this appendix, we prove the opposite-parity structure of the Cayley–Klein elements stated in Eq. (8). The proof does not rely on palindromic symmetry and therefore applies to an arbitrary sequence of identical nominal π pulses with independently programmed phases.

Define

$$Z = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (\text{F1})$$

The primitive propagator introduced in Eq. (7) is

$$U_\phi^{(\pi)}(q, \alpha) = \begin{pmatrix} qe^{i\alpha} & -i\sqrt{1-q^2}e^{i\phi} \\ -i\sqrt{1-q^2}e^{-i\phi} & qe^{-i\alpha} \end{pmatrix}. \quad (\text{F2})$$

Although q represents a physical failure amplitude, it may be treated as a formal expansion variable near $q = 0$. The quantity $\sqrt{1-q^2}$ is an even analytic function of q . Therefore, the transformation $q \rightarrow -q$ reverses the sign of the diagonal matrix elements of Eq. (F2), while leaving the off-diagonal elements unchanged. It follows that

$$U_\phi^{(\pi)}(-q, \alpha) = -ZU_\phi^{(\pi)}(q, \alpha)Z. \quad (\text{F3})$$

Consider now an N -pulse composite sequence,

$$U^{(N)}(q, \alpha) = U_{\phi_N}^{(\pi)}(q, \alpha) \cdots U_{\phi_2}^{(\pi)}(q, \alpha) U_{\phi_1}^{(\pi)}(q, \alpha). \quad (\text{F4})$$

For compactness, let $U_k = U_{\phi_k}^{(\pi)}(q, \alpha)$. Applying Eq. (F3) to every constituent pulse gives

$$\begin{aligned} U^{(N)}(-q, \alpha) &= (-1)^N (ZU_N Z) \cdots (ZU_1 Z) \\ &= (-1)^N ZU_N (Z^2) \cdots (Z^2)U_1 Z \\ &= (-1)^N ZU^{(N)}(q, \alpha)Z, \end{aligned} \quad (\text{F5})$$

where all intermediate factors cancel because $Z^2 = I$.

The composite propagator has the Cayley–Klein form

$$U^{(N)}(q, \alpha) = \begin{pmatrix} A^{(N)}(q, \alpha) & B^{(N)}(q, \alpha) \\ -B^{(N)*}(q, \alpha) & A^{(N)*}(q, \alpha) \end{pmatrix}. \quad (\text{F6})$$

Conjugation by Z gives

$$ZU^{(N)}(q, \alpha)Z = \begin{pmatrix} A^{(N)}(q, \alpha) & -B^{(N)}(q, \alpha) \\ B^{(N)*}(q, \alpha) & A^{(N)*}(q, \alpha) \end{pmatrix}. \quad (\text{F7})$$

Combining Eqs. (F5) and (F7), and comparing the corresponding matrix elements, yields

$$A^{(N)}(-q, \alpha) = (-1)^N A^{(N)}(q, \alpha), \quad (\text{F8a})$$

$$B^{(N)}(-q, \alpha) = (-1)^{N+1} B^{(N)}(q, \alpha). \quad (\text{F8b})$$

For an odd number of pulses, $N = 2m + 1$, these relations reduce to

$$A^{(2m+1)}(-q, \alpha) = -A^{(2m+1)}(q, \alpha), \quad (\text{F9a})$$

$$B^{(2m+1)}(-q, \alpha) = B^{(2m+1)}(q, \alpha). \quad (\text{F9b})$$

Thus, $A^{(2m+1)}$ is an odd function of q , whereas $B^{(2m+1)}$ is an even function of q .

Since the constituent propagators are analytic around $q = 0$, the corresponding Taylor expansions contain only the powers allowed by these parity relations:

$$A^{(2m+1)}(q, \alpha) = qA_1(\alpha) + q^3A_3(\alpha) + q^5A_5(\alpha) + \cdots, \quad (\text{F10a})$$

$$B^{(2m+1)}(q, \alpha) = B_0(\alpha) + q^2B_2(\alpha) + q^4B_4(\alpha) + q^6B_6(\alpha) + \cdots. \quad (\text{F10b})$$

The coefficient functions $A_{2j+1}(\alpha)$ and $B_{2j}(\alpha)$ also depend implicitly on the programmed phases $\phi_1, \dots, \phi_N$.

Equation (F10) proves Eq. (8). It also explains the alternating cancellation hierarchy for constant X gates.

The condition $A_1 = 0$ removes the leading diagonal error, $B_2 = 0$ removes the next off-diagonal correction, $A_3 = 0$ removes the following diagonal correction, and so forth.

For completeness, when $N = 2m$ is even, Eq. (F8) gives

$$A^{(2m)}(-q, \alpha) = A^{(2m)}(q, \alpha), \quad (\text{F11a})$$

$$B^{(2m)}(-q, \alpha) = -B^{(2m)}(q, \alpha). \quad (\text{F11b})$$

Hence, for even-length sequences, $A^{(2m)}$ contains only even powers of q , while $B^{(2m)}$ contains only odd powers. The parity structure in Eq. (8) is therefore determined solely by the pulse-number parity and does not require any additional symmetry of the phase sequence.

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